CAE

OBJECTIVES

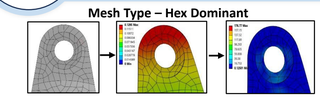
- Virtual validation of BAJA vehicle components under real operating loads
- Assessment of structural safety, durability, and vibration
- Weight optimization while maintaining required strength
- Minimization of physical through CAE driven design

MESH TYPE

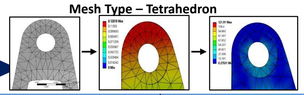
- Mesh quality verified using skewness, aspect ratio, warpage, Jacobian, and non-orthogonality limits
- Hex-dominant mesh delivered superior element quality and smoother stress distribution than tetra mesh
- Improved accuracy and faster convergence with fewer distorted elements

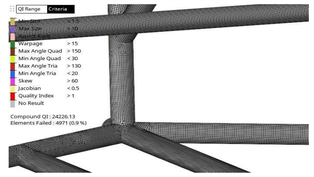
FEATURES AND IMPROVEMENTS

Mesh Type – Hex Dominant



Mesh Type – Tetrahedron





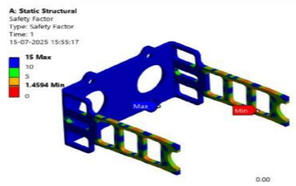
PARAMETERS	VALUE	PARAMETERS	VALUE
Skewness	60	Jacobian	0.60
Aspect ratio	5	Non-orthogonally	45
Warpage	5	Aspect ratio	6

LINEAR VERSUS NON-LINEAR ANALYSIS

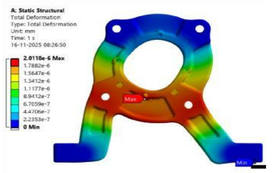
No.	Material	Material Properties	Geometry	Contact Influence	Overall Classification
1	Al Series 6065	Linear elastic below yield	Linear unless large deformation	Contact does not affect material linearity	Linear
2	AL 7075	Linear elastic below yield	Linear unless large deformation	Contact does not affect material linearity	Linear
3	EN 36	Linear elastic	Generally linear	Linear	Linear
4	EN 19	Linear elastic	Linear	Linear	Linear
5	Mild Steel (MS)	Linear elastic	Linear	Linear	Linear
6	SS 410	Linear elastic	Linear	Linear	Linear
7	AISI 4130	Linear elastic	Linear unless slender geometry buckles	Linear	Linear

STATIC ANALYSIS

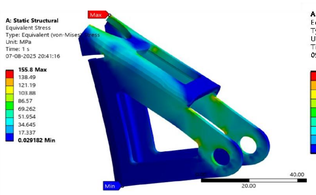
Mc mount – mild steel



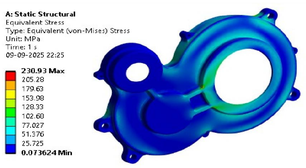
Motor Mount – AL 7075 T6



Front upper damper mount



Gearbox Casing – Al 7075 T6

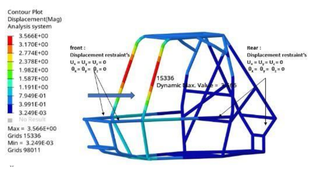


RESULT TABLE

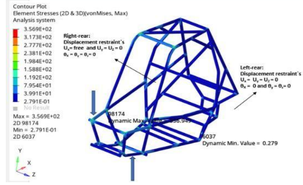
Component	Stress	Deformation	FOS
Motor mount	249.93 Mpa	0.020143 mm	1.62
Front damper mount	155.88 Mpa	0.026504 mm	1.60
Gear box casing	230.93 MPa	0.5959 mm	1.12

ROLLCAGE ANALYSIS

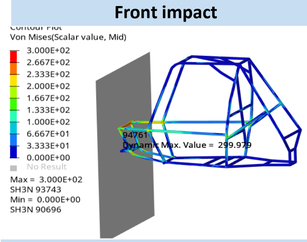
Pitch over analysis



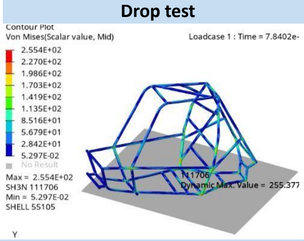
Torsional



Front impact



Drop test

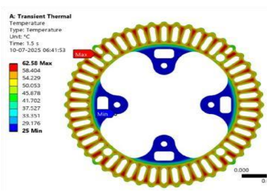


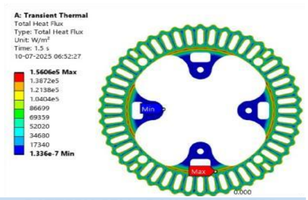
RESULT TABLE

Impact	Roll Over	Torsional	Front	Drop
Load	7.34 KN	9.16 KN	16.66 m/s	9.81 m/s²
Stress	424.77 MPa	356.94 MPa	299.9 MPa	255.37 MPa
Deform.	3.56 mm	6.18 mm	106.543 mm	97.23 mm
Fos	1.53	1.82	2.17	2.54

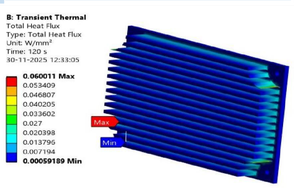
THERMAL ANALYSIS

BRAKE ROTOR - SS410

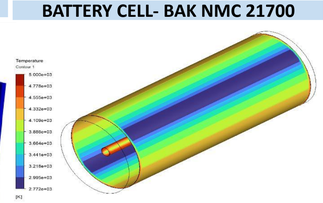




HEAT SINK - Al 6065 T6



BATTERY CELL- BAK NMC 21700

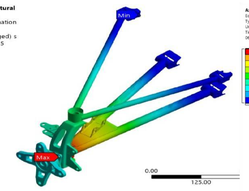


MAX STRESS AND MAX DEFORMATION

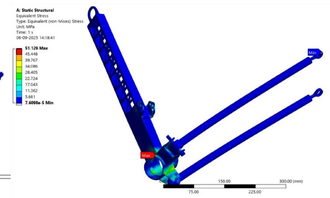
Component	Temp rise	Heat flux
Front Rotor – SS410	62.58°C	1.56e5 W/mm²
Heat Sink – Al 6065 T6	58°C	0.06011 W/mm²
Cell – BAK NMC 21700	50 °C	0.06011 W/mm²

DYNAMIC ANALYSIS

FRONT ASSEMBLY



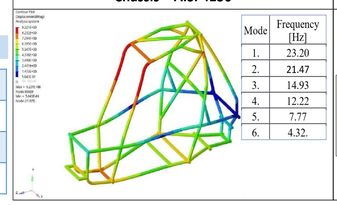
REAR ASSEMBLY



Component	Stress	Deformation	FOS
Front assembly	85.48 MPa	1.40 mm	3.72
Rear assembly	51.129 MPa	0.0568065 mm	4.88

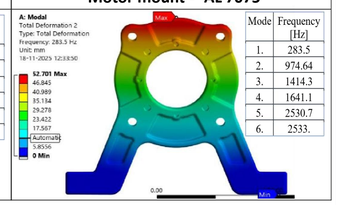
MODAL ANALYSIS

Chassis – AISI 4130



Mode	Frequency [Hz]
1.	23.20
2.	21.47
3.	14.93
4.	12.22
5.	7.77
6.	4.32.

Motor mount - AL 7075

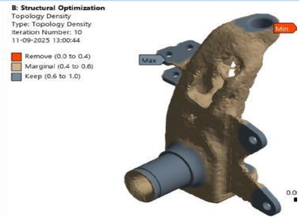


Mode	Frequency [Hz]
1.	283.5
2.	974.64
3.	1414.3
4.	1641.1
5.	2530.7
6.	2533.

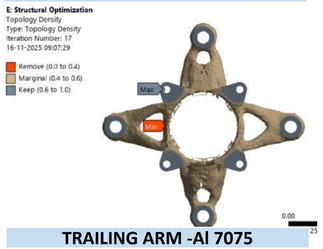
TOPOLOGY ANALYSIS

s²

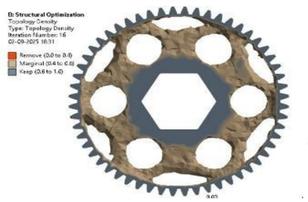
KNUCKLE -Al 7075



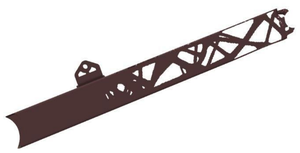
WHEEL HUB -Al 7075



Gear – EN 36

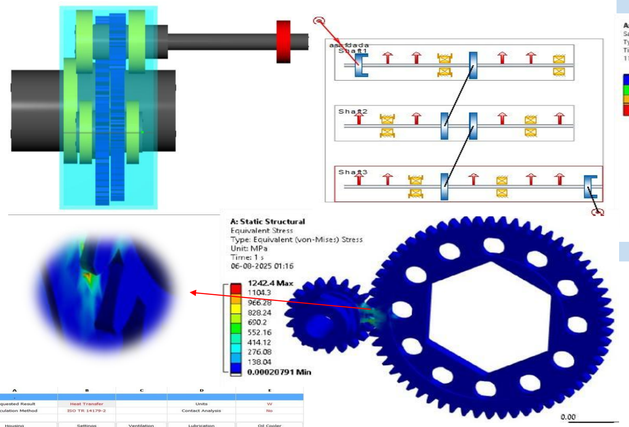


TRAILING ARM -Al 7075



component	KNUCKLE	WHEEL HUB	GEAR	TRAILING ARM
Wt before topology	1.8244 kg	0.89258 kg	0.3616 kg	0.73098 kg
Wt After topology	1.4835 kg	0.51762 kg	0.2656 kg	0.48499 kg

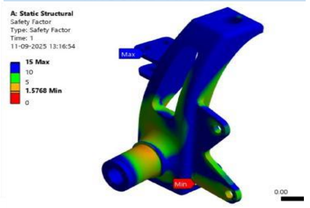
POWERTRAIN ANALYSIS



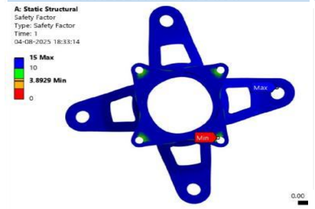
Material	*EN 19*
Stress	**1055.6 MPa**
Deformation	0.19607 mm
FOS	1.69

FATIGUE ANALYSIS

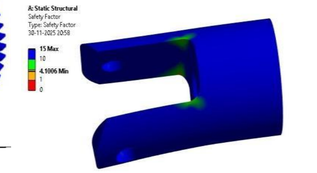
KNUCKLE -Al 7075



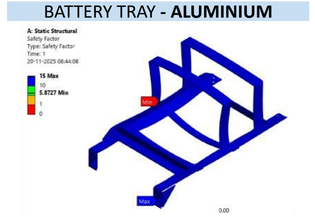
WHEEL HUB -Al 7075



CLEVIS – EN 19



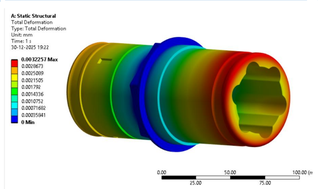
BATTERY TRAY - ALUMINIUM



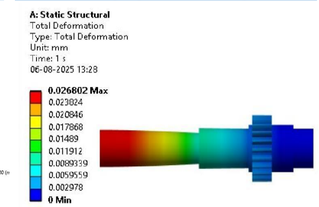
component	knuckle	wheel hub	clevis	battery tray
stress	319.01 MPa	156.27 MPa	112.18 MPa	108.27 MPa
fos	3.89	1.57	4.10	5.87
life				

TORSIONAL ANALYSIS

SPOOL – EN 19

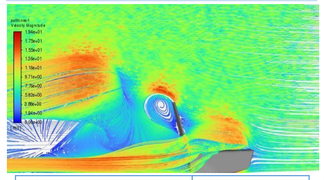


INPUT SHAFT – EN 19



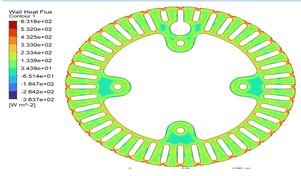
CFD ANALYSIS

CHASSIS



inlet	***16.67 m/s***
Turbulence model	*k–ω SST*
y+ target (k–ε)	30 - 300
Drag obtained	90 – 92 N

BRAKE ROTOR



braking condition	
Turbulence model	*k–ω SST*
Ambient Temp	27 °C
Max Temp	73 °C